

Statistical Models in Computational Biology

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Problem 22: d-separation

(3 points)

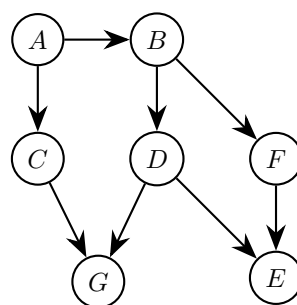


Figure 1: DAG

For the following Bayesian network,

- (i) Write down all the variables that are d-separated from A given {C,E}.
- (ii) Indicate whether each statement is true or false and explain your choice.
 - i) C is independent of D.
 - ii) B is conditionally independent of C given {F, G}.
 - iii) G is conditionally independent of E given {A, D}.
 - iv) C is conditionally independent of F given {G, E}.
 - v) C is conditionally independent of E given its Markov blanket (of C).

Problem 23: Structure Learning in R

You are given a data set containing multivariate normally distributed data with a dependency structure corresponding to the DAG in Figure 1. We will use the PC algorithm for structure learning, but first we will look at the respective steps involved in the inference procedure.

Testing for marginal and partial correlations:

Note that for a multivariate normal distribution, uncorrelated variables are independent. However, it is important to keep in mind that this implication does not hold in general.

We can define the correlation coefficient as

$$\rho_{X,Y} := \frac{Cov(X,Y)}{\sqrt{Var(X)}\sqrt{Var(Y)}} \quad (1)$$

for which in R, we can use the function `cor.test()` to test the null hypothesis of no correlation. Similarly, we can define the partial correlation coefficient as

$$\rho_{X,Y|Z} = \frac{\rho_{X,Y} - \rho_{X,Z}\rho_{Y,Z}}{\sqrt{(1 - \rho_{X,Z}^2)}\sqrt{(1 - \rho_{Y,Z}^2)}} \quad (2)$$

The partial correlation $\rho_{X,Y|Z}$ equals the correlation between residuals from the linear regressions of X on Z , and Y on Z , respectively.

Hence, we can compute the partial correlation of e.g., $\rho_{A,B|C}$ as follows:

- Linearly regress X on Z (that is, with X as the response variable and Z as the explanatory variable). Compute and store the residuals.
- Linearly regress Y on Z . Compute and store the residuals.
- Use the function `cor.test()` to test the null hypothesis of no correlation between the residuals of X (regressed on Z) and the residuals of Y (regressed on Z).

Re-evaluate the independence relations (via correlation tests) from Problem 22 given the provided data (3 points)

Run the PC algorithm (2 points)

Install and load the R package `pcalg`. Use the function `pc()` to run the PC algorithm on the data, and plot the result (use $\alpha = 0.05$ as the significance level). For hints, see footnote¹.

Run the order MCMC with BGe Score (1 point)

Install and load the R package `BiDAG`. Initialize the parameters with `Score <- scoreparameters("bge", data)` on the data using the Bayesian Gaussian equivalent (BGe) score. Run the iterative MCMC algorithm with `maxBN <- learnBN(Score, algorithm = "orderIter")` to learn the maximum scoring DAG and plot its equivalence class CPDAG.

Compare the two graph estimates (1 point)

¹If you encounter trouble installing `pcalg`, make sure that you have a recent version of R, such as 3.4.4 or higher. For the PC algorithm applied to normally distributed data, the sufficient statistics are the sample correlation matrix C of the data (see `cor()`), as well as the sample size n . Supply these as a list for the `suffStat` argument of the function `pc()`. Specify `indepTest = gaussCITest`, and set a reasonable significance level `alpha` for the independence tests. Supply the node names `colnames(data)` to the argument `labels`. Note that when plotting a pDAG, undirected edges are drawn as ' $\leftrightarrow$ ' rather than ' $-$ '.